

# Not So Surelock



**A STEM Day Presentation About Forensic Science  
Covering DNA Analysis, Creativity, And Problem  
Solving Skills.**

# Not So Surelock Overview



## STEM Day

STEM day is a free event facilitated by the Center for Initiatives in STEM and has now become a University of Central Florida tradition. Hosted every spring and fall, this event aims to teach elementary and middle school students about STEM topics through immersive presentations, exhibits, and activities. Brought together by UCF students and faculty, STEM day helps young minds foster an interest in these fields.

Not So Surelock is a project created by SPC1603 honors students for STEM Day, centered on forensic science and tailored for a young audience. This project created an opportunity for our team to employ the presentation skills learned in class such as content tailoring and effective communication strategies. At this exhibition, students will become detectives and gather evidence to find solve the case of the lost Mona Lisa.

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This portfolio serves as a comprehensive documentation of our team's journey in bringing the Not So Surelock project to fruition.

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# INTRODUCING OUR TEAM



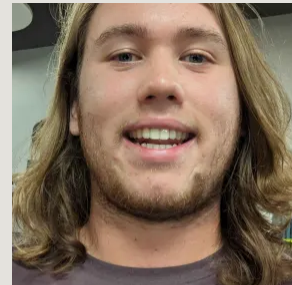
**Tahel Peretz**

Hi, I'm Tahel! I'm studying forensic science at UCF and I am a member of the Burnett Honors College. In the future, I hope to work as a patent attorney in big law.



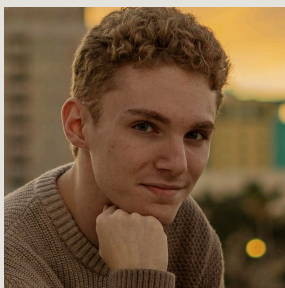
**Sarayu Marri**

Hi! I'm Sarayu, a computer science major and digital media minor at UCF. I'm a member of the Burnett Honors College and hope to pursue video game development in the future.



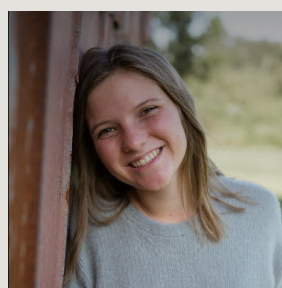
**Colin McLallen**

Hello. I'm a second-year computer science major at the Burnett Honors College in UCF. Whenever I'm not working on my classes or playing video games, I enjoy marching trumpet in the Marching Knights.



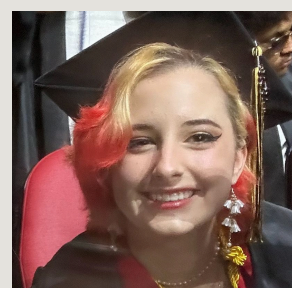
**Derek Felice**

Hi, my name is Derek. I am a first year studying astrophysics and a member of the Burnett Honors College at UCF. After I graduate, I hope to work for NASA.



**Avery Hanebrink**

Hi, my name is Avery! I'm studying mechanical engineering at the Burnett Honors college at UCF. I hope to become an engineer for a theme park after I graduate.



**Vanessa Flores**

Hello, my name is Vanessa Flores! I am majoring in molecular and cellular biology and minoring in psychology. I'm pursuing molecular pharmacology.

# Official Pitch



## The Center for Initiatives in STEM

### Fall STEM Day 2024 Exhibitor Information

**Date of Event:** Friday, November 1, 2024

**Event Location:** UCF Main Campus

**Deadline:** Return by Friday, *September 13* to [Renee.Johnston@ucf.edu](mailto:Renee.Johnston@ucf.edu)

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#### Exhibitor Information:

Name: Tahel Peretz

Professional Title: Communication Representative - SPC 1603H

Department/Organization: Burnett Honors College

Office phone: N/A Cell phone: \_\_\_\_\_

Email: \_\_\_\_\_

**Title of activity** (should be short & catchy):

Not So Surelocks: Forensic Science Investigation

**Description of interactive exhibit/activity.** What will you demonstrate and how will you engage visitors? Avoid vague descriptions as well as descriptions that use technical jargon. This description will be used in printed/online materials:

With a twist on the typical escape room, detectives will work to solve a criminal investigation as the time ticks down. Through the Not So Surelocks interactive learning experience, students will act as detectives and will be tasked with finding physical evidence and conducting various types of DNA analysis to discover which of the six suspects committed the art heist. Before the forensics investigation, multiple witnesses will explain what happened at the crime scene, and investigators

**What age group do you expect to engage in your activity?** (choose all that apply)

☒ Elementary school    ☒ Middle school    \_\_\_\_\_ High school

Sessions will begin every 45 minutes for a total of 5 sessions (lasting 30-35 min). How many times will you offer the event throughout the day? 5

If you are not offering the event for all 5 sessions, please indicate which times your event will be offered:  
N/A

Do we need to help you find a space for your event? Yes

If yes, do you have any specific needs? (desks, projector) Multiple tables, room that gets dark, projector

If you have your own space, tell us the location of your event (building, room #): N/A

How many students can attend each offering of the event?: 25

Please provide a "Day Of" contact name and phone number: \_\_\_\_\_

# Pitch Process



This idea was pitched with the goal to teach students about the scientific method in a thought-provoking and engaging way. Tahel Peretz, our communication representative, pitched the concept as she is a forensic science major passionate about the subject. It was then fine tuned by our members to cover different types of evidence that would be found in crime scenes, such as fingerprints, CCTV footage, and cross sampling of dirt (in this case paint).

To add a unique spin to the exhibition, the room was decorated as an art museum, giving students an initiative with a story. Their goal was to examine the theft of a painting that was “stolen” by one of our members. In this way students will use critical thinking as they become detectives to solve the case.

Our design was then altered so each member was responsible for a “station” that covers one type of evidence as students follow a trail of clues. This prevented one person from receiving too much “traffic” and spread out supervision in the room. However, some of our station ideas such as UV light detection and tool mark investigation were cut due to time constraints.

## **Stations Discussed in Detail:**

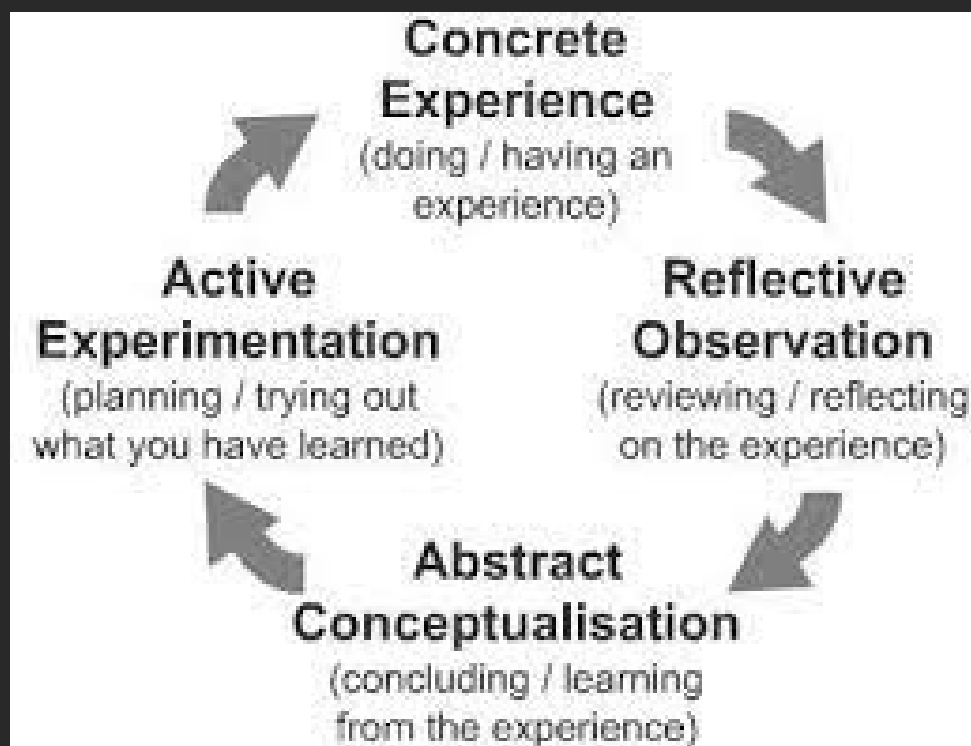
- **Paint Splatters:** As the suspect stole the painting, they knocked over a can of paint, leaving evidence of the direction they headed and how the item was stolen. Derek Felice, who was studying physics, volunteered to explain the types of splatters and their relation to how high the paint can was displayed.
- **Fingerprints:** This station allows students to examine the different types of ridges in one’s finger and use this information to track the suspect down. Sarayu Marri suggested having participants make a stamp of their own fingerprint to reinforce this idea in a tactile way.
- **Footprints:** Vanessa Flores learned about the correlation of one’s footprint to their height in a high school forensics class. This idea could help students determine a suspect based on the heights of our members.
- **CCTV camera footage:** This station would directly tie into the technological advancements in modern forensics. However, our group was concerned this would make the activity too simple for students.

# Audience Analysis



Our research taught us that a “hands-on,” student-led approach was the most effective method of teaching a younger audience. For this reason, we ensured to include tangible elements in our exhibition and provided opportunities for students to use deductive thinking themselves. This allowed them to explore different methods of thinking in the field on their own.

One example of such tangible elements was the fingerprint station where students discovered traces left behind by the art thief while making a stamp of their own fingerprint as a take-home memory. Following this trend, our presentation flowed around the room and allowed students to interact with physical evidence rather than sitting through a PowerPoint. At the end of our activity, students voted on who they believed had stolen the painting, and the culprit was revealed through a clip of the museum’s CCTV footage (this will be discussed again later, as it was the compromise we found on keeping that station).



Leggett, University of Newcastle

## Reflection on Analysis

This process was the most challenging part of our project as much of the techniques we were referencing were from actual crime scenes. Thus, we had to find a balance between what was appropriate for young students to handle and what would be intellectually stimulating for them.



## STEM DAY Outline

### Small Talk:

Welcome Message: While you all were out trick-or-treating last night, a crime occurred at the local art museum and we need your help to find out who stole the Mona Lisa.

**Rules/Procedures:** In this activity, you all will act as detectives at the crime scene and your job will be to use the evidence that you find at the crime scene to figure out who the true culprit is. Right now, we have six suspects that we are considering and the evidence at the crime scene that you will look through will help eliminate suspects. At each of the four stations, you will be conducting an activity with four different types of evidence, namely, physical evidence, paint splatter, fingerprints, and footprints and you will learn how they each play an important role in determining the culprit. After we look at all the pieces of evidence, we will use our analysis of the evidence and narrow down the suspects and solve our case.

**Attention Getter:** The Mona Lisa has been stolen and we need to figure out who the culprit is! Let's watch a video of what the witnesses have said at the crime scene so we can gain a better understanding of what happened since none of us were actually there.

### Credibility:

My name is Tahel and I am a Forensic Science major interested in learning about the way we can use science and apply it to our legal system.

My name is Vanessa Flores and I am a Pre-Health major. In High School, I took a Forensics course and participated in a staged crime scene investigation much like this one. I had a lot of fun but more importantly, learned how forensics can be applied to almost anything at the scene of the crime.

My name is Derek and I am majoring in Astrophysics. My mom is a lawyer so I have always had an interest in law, criminal investigations, and the science behind them.

My name is Avery and I'm majoring in Mechanical Engineering. I love crime shows, so I am very interested in the forensic science process and figuring out who committed the crime.

My name is Sarayu and I'm majoring in computer science and minoring in digital media. I hope to get into game development with mystery games being my favorite.

My name is Colin and I'm majoring in computer science. Forensic science, specifically the problem-solving aspect, has always been interesting to me. Because of this, I want to get others interested as well.

**Relevance:** Crimes are happening every day all around us and while they may not affect our daily lives, we cannot just ignore them. Performing a crime scene investigation will help you learn skills in data collection, observation, analysis, and problem-solving which will make you a more creative thinker. Maybe this activity will even inspire you to want to work in the crime field in the future!

**Thesis:** Aspects like observing evidence from an environment, physical and life sciences, and a little bit of math all come together to form forensic science, which law enforcement uses to identify criminals and keep us safe.

**Preview:** Today, we will gather physical evidence found at a crime scene, analyze paint splatters, collect and compare fingerprints, and calculate the height of a person from their footprints to figure out what happened at the crime scene and determine the true perpetrator.

- Claim Physical Evidence
  - Evidence - We found hair, beads, feathers, and other observable and tangible evidence left behind by the culprit at the crime scene.
  - Personal connection - Everywhere you go you leave physical evidence behind. Have you ever left something behind and not realized it? That is physical evidence, and it can be found by other people. We can use physical evidence to try and narrow down what a person was wearing or where they might have moved through a crime scene.
  - Audience relevance - For this station, our job is to use these pieces of physical evidence to narrow down who the culprit might be. We will also use pieces of physical evidence such as knocked-over paintbrushes or spilled paint to figure out how they might have committed the crime.
- Claim Spatter
  - Evidence - At a crime scene, we can often find other types of evidence that was left behind, such as a paint spatter found from the painting of the fake Mona Lisa. By looking into how the paint was splattered at the scene, whether it is from a farther distance, a closer distance, or the angle from

# Official Plan Cont.



which it was spattered, we can better tell the sequence of events that occurred at the crime. This is integral as it will help eliminate eyewitnesses and suspects that had nothing to do with the crime.

- Personal connection - All of us have spilled something before, even if we don't remember it. Maybe it wasn't paint, it could have been juice, water, milk, anything. You probably didn't think about this then, but that spill left a spatter pattern, which could be used to trace the spill back to you.
- Audience relevance - For this station, we need to analyze the paint spatters that were left at the crime scene. This will help us find out what happened last night, and we can help eliminate some innocent people from our suspect list. First, we're going to see how dry the paint is to figure out how recently the crime occurred. Then we will look at the shape and size of the paint spatters. We have a chart that shows how to tell what angle and height the paint came from, so we will match up the spatters from the crime scene to figure out what happened.
- Claim Fingerprints
  - Evidence - Fingerprint analysis involves comparing fingerprints found at the scene of the crime with fingerprints of known individuals and assessing how similar they are. An examiner compares details like the shapes the ridges on your finger form, and where the lines start or end.
  - Personal connection - A lot of the phones that we use have a fingerprint scanner that takes in your fingerprint. These scanners either use a picture of your fingerprint, an electrical map of your fingerprint, or an ultrasonic pulse similar to what bats use to map your fingerprint.
  - Audience relevance - In one of the activities, we will compare fingerprints found at the scene of the crime with known fingerprints, as well as take our own fingerprints.
- Claim Footprints
  - Evidence- A shoe print within spilled paint is found at the scene of the crime. We took a picture and made copies for you to analyze.
  - Personal connection- Before we start analyzing the evidence I want to ask some of you how tall you are. (person answers) well using that I know that your footprint is (insert using chart). All of us live in Florida and I'm sure most of you have experienced walking through your front door only to realize the endless rain made it muddy and now you have a trail of footprints following you. Well this same thing can happen during a crime. In the case of our current crime scene, the culprit spilled paint and walked through it leaving their prints behind. Just like how I was able to know your foot size based on your height, we can determine the criminal's height using their footprint.
  - Audience relevance- For this station it is our job to analyze the shoeprint to the suspects. Shoe size can tell us the length of a person's foot and forensic anthropologists (bone scientists) found that foot length is the same as about 15% of a person's height. This means that if we know the length of the shoe we can divide that measurement by 0.15 and we know the person's height. Now, I know that's a lot of math so we provided a chart with a list of shoe sizes and what height that would relate too. It is our job to measure the print and compare using the chart to later use when finding our suspect.

**Review:** Today we gathered physical evidence from a crime scene, analyzed paint splatters, collected and compared fingerprints, and calculated height from gathered footprints all to put together an idea of what happened at the crime scene.

**Thesis restatement:** Aspects like observing evidence from an environment, physical and life sciences, and a little bit of math all come together to form forensic science, which law enforcement use to identify criminals and keep us safe.

**Clincher:** Numerous aspects of STEM come together to form many different subjects, like forensic science. Whatever you are interested in, STEM is involved in some way or another, so keep pursuing STEM.

**Parting Message:** Thank you all for coming to UCF's Stem Day. We hope that many of you will become STEM majors at UCF.

**Plan for material/activity accessibility:** Our activity contains multiple stations so if someone can not participate in one we can make sure another station fits.

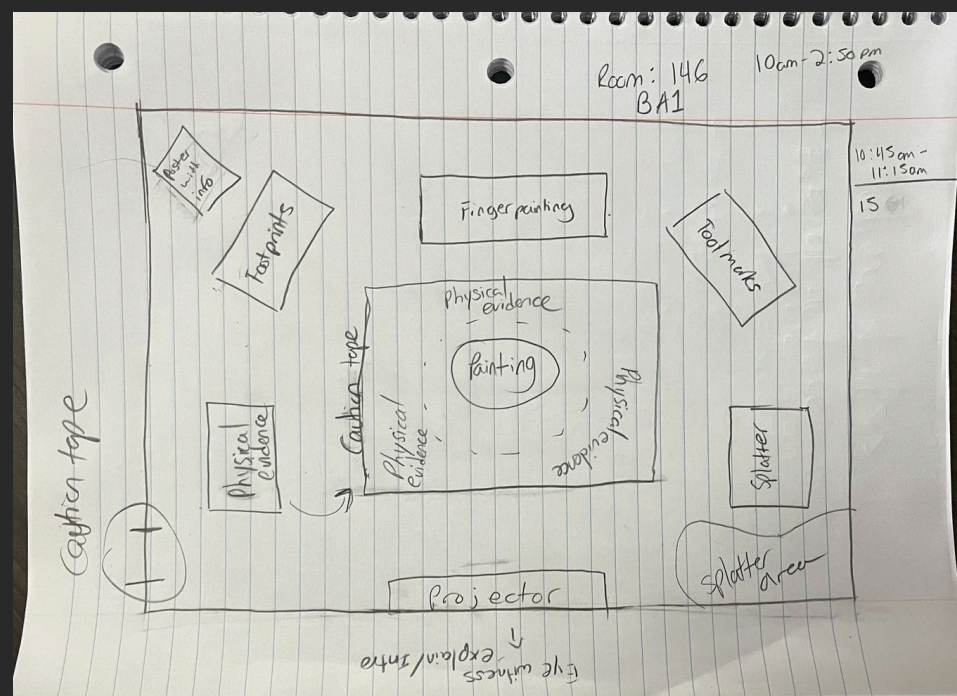
# Worksheet/ Layout



We created this worksheet to help guide students around the room and collect the evidence they find.

| <u>NOT SO SURELOCK</u>                                                                                   |                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DETECTIVE NAME: _____<br>CLASS: _____<br>DATE: 11-1-2024                                                 |                                                                                                                                                                                  |
| <b>OBJECTIVE:</b>                                                                                        | On Halloween night, a mysterious robbery took place at the UCF Art Museum, and the famous Mona Lisa was stolen! Join Sherlock in solving the case and help track down the thief! |
| <b>FOOTPRINT:</b>                                                                                        | • Shoe Length: _____<br>• Height: _____                                                                                                                                          |
|                                                                                                          | <b>FINGERPRINT:</b><br>• Type of Fingerprint: _____<br>• Suspect Match: _____                                                                                                    |
|                                                                                                          | <b>PAINT SPLATTER:</b><br>• Angle: _____<br>• Pattern: _____                                                                                                                     |
| <b>WHO DID IT?</b><br>Suspect A      Suspect C      Suspect E<br>Suspect B      Suspect D      Suspect F |                                                                                                                                                                                  |

| <u>PHYSICAL EVIDENCE:</u> |  |
|---------------------------|--|
| <b>EVIDENCE 1:</b>        |  |
| <b>EVIDENCE 2:</b>        |  |
| <b>EVIDENCE 3:</b>        |  |
| <b>EVIDENCE 4:</b>        |  |
| <b>EVIDENCE 5:</b>        |  |
| <b>EVIDENCE 6:</b>        |  |
| <b>EVIDENCE 7:</b>        |  |
| <b>EVIDENCE 8:</b>        |  |



This was our initial plan for the layout of the room, done by Vanessa. We wanted the "crime scene" sectioned off with caution tape in the center of the room. Students would begin at the crime scene and follow the clues around the room. The trail of evidence would end at the backdoor of the classroom where it assumed the culprit "escaped" from.

# Lesson Plan Analysis



Ultimately, we decided to keep the previously shown layout of the room. Students would begin at the center of the room where they can deduce what occurred at the crime scene. Then, they would be split into groups and rotate around the room as they gather evidence on who committed the crime. Each of our group members would be stationed at an activity. We would regroup at the back exit of the room after the students have gone through each station. This is where we would ask them to guess who the culprit of the crime was, and then show them the CCTV camera footage revealing the answer. The CCTV was only shown at the end to maintain the air of mystery.

The lesson plan was the most time consuming portion of our project as it was the bulk of our exhibition. Although it overshadowed our initial presentation, we felt this was necessary for students to actively participate and recall these topics later on. The lesson plan also allowed each of our members time with the students, which helped people-orient our exhibition and dive into the nuances of the subject based on the questions asked.



## Presentation Planning

| Height    | Shoe Length | Height    | Shoe Length  |
|-----------|-------------|-----------|--------------|
| 48 inches | 7.7 inches  | 60 inches | 9.5 inches   |
| 49 inches | 7.8 inches  | 61 inches | 9.6 inches   |
| 50 inches | 8 inches    | 62 inches | 9.8 inches   |
| 51 inches | 8.15 inches | 63 inches | 9.95 inches  |
| 52 inches | 8.3 inches  | 64 inches | 10.1 inches  |
| 53 inches | 8.45 inches | 65 inches | 10.25 inches |
| 54 inches | 8.6 inches  | 66 inches | 10.4 inches  |
| 55 inches | 8.75 inches | 67 inches | 10.55 inches |
| 56 inches | 8.9 inches  | 68 inches | 10.7 inches  |
| 57 inches | 9.05 inches | 69 inches | 10.85 inches |
| 58 inches | 9.2 inches  | 70 inches | 11 inches    |
| 59 inches | 9.35 inches | 71 inches | 11.15 inches |

This poster is part of the footprints station, providing the suspect's heights.

Our presentation consisted of a short slideshow at the beginning of the event explaining what forensic science is, its importance, and a list skills important to forensic scientists. We then introduced the crime with a interview between a detective and security guard who was present. Created by Tahel's roommates, this video helped set the atmosphere of our activity. Furthermore, Avery Hanebrink came up with the idea to cover over the crime scene with a tarp as we were presenting. This would prevent students from getting distracted by the props before they understood the activity. This idea also allowed us to do a "reveal" of the missing painting that would catch their attention.

Overall, the presentation was an engaging process that our entire team contributed to. Following our lesson plan, it was enjoyable and fairly simple to execute.

# Game Day (Highs and Lows)



## Set Up



The set up process went smoothly after our careful planning. However, we adjusted our presentation to student input after the initial groups. This was done in the 15 minute intervals between groups, where our team regrouped to discuss the highs and lows of our exhibition. We also found differences in how it was recieved between different age groups.

Here were the highs and lows we faced with each group:

### **Group 1 (10:45am - 11:15am)**

Highs:

- youngest group yet caught onto the lesson plan the fastest

Lows:

- students grew upset if our physical qualities (ex. fingerprint, height) did not match with the suspects; accounted for this by not giving participants this information
- difficult to direct students to each station

### **Group 2 (11:30am - 12:00pm)**

Highs:

- students became interested in our set up/asked about building our project

Lows:

- difficult to manage different grade levels in same group



# Game Day (Highs and Lows)



## Group 3 (12:15pm - 12:45pm)

### Highs:

- smallest and most engaged group
- students were older and interested in our majors
- able to adapt to an older age group

### Lows:

- lesson felt too simple/finished quickly

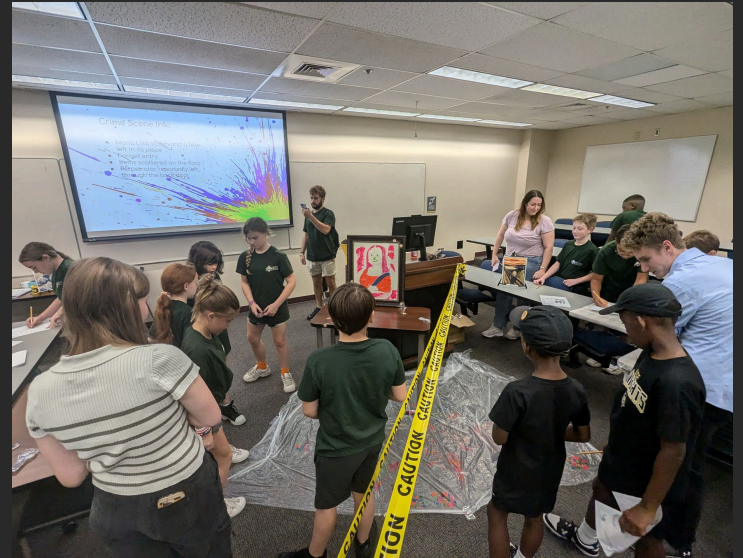
## Group 4 (1:45pm - 2:15pm)

### Highs:

- students spoke about their own interest in STEM

### Cons:

- was after lunch and students had low energy



## Reflection

Overall, we were happy with the student engagement displayed. Many students asked further questions about the topic or were able to figure out the intent behind our activity before it was revealed. This meant that our message design changed throughout the day to give us more time to converse about the field on a deeper level. This was especially present with the groups of middle schoolers. An example to how we adapted our lesson plan was Colin McLallen identifying students who played instrumnets or a sport based on their fingerprint. This increased their interest in a station that would otherwise have been too simple.

## Takeaways

Thus, one of our biggest takeaways and advice for future STEM Day groups is to keep the exhibition flexible. Looking back, we wish we had known the age groups and backgrounds of our participants and we had to account for this by quickly adapting to input on game day. A flexible lesson plan accounts for this by providing activities for both multiple age groups. We were surprised by the students' quick wit and found the best way to appeal to them was by having a people-oreinted presentation. Our presentation accounted for this and made the experience interactive, with many students making conversation with our members about our own fields of study. They appered greatly interested in how the STEM field presented could interact with other fields outside of STEM.

# Initiatives in STEM



Our project could not have been made possible without STEM Day's facilitator, Initiatives in STEM (iSTEM). iSTEM was created with the mission to promote education and research in STEM fields within the university and among the community. STEM Day is merely one example of its reach to the community. Additionally, we would like to extend our gratitude to Renee Johnston and Dr. Adam Parrish for providing us with this opportunity and supporting our ideas into fruition.

Currently iSTEM has presence in the College of Engineering and Computer Science, College of Sciences, Colleges of Medicine, Optics and Photonics, and Community Innovation and Education.

